Dineth Sanjula Jayakody

Ph.D. Student in Computer Science, Old Dominion University (ODU), USA
B.Sc. (Hons) in Electrical and Information Engineering (First Class Honours), University of Ruhuna, Sri Lanka
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Summary

Motivated and self-driven Electrical and Information Engineering graduate specialized in Artificial Intelligence (AI) and Machine Learning (ML). Passionate about the transformative potential of AI and ML to drive innovation and committed to academic excellence. Possesses a keen interest in research, eager to explore and contribute to cutting-edge advancements in the field. Excels in both academic pursuits and sports, demonstrating a balanced approach to personal and professional development. Recognized for a tenacious work ethic, proactive mindset, and readiness to tackle complex challenges head-on.

Education

Old Dominion University — Norfolk, Virginia, USA
Ph.D. student in Computer Science — *2025 - present*

University of Ruhuna — Galle, SriLanka
B.Sc.(Hons) Electrical and Information Engineering - OGPA 3.94/4.00 (Highest in the batch) — *2021 - 2025*

Vellore Institute of Technology — Chennai, India
Certificate in AR/VR and Robotics — *2024*

E-soft Metro Campus — Colombo, SriLanka
Diploma in Information Technology — *2019 - 2020*

G.C.E. Advanced Level Examination — Sri Lanka
2 A's 1 B and selected to study engineering — *2018*

G.C.E. Ordinary Level Examination — Sri Lanka
9 A's and ranked 7 th in Sri Lanka among 350,000 students — *2015*

Work Experience

Graduate Research Assistant — August 2025 - present
Department of Computer Science - Old Dominion University — *Virgina, USA*

- Collaborating in a cross-disciplinary lab environment (Computer Science, Data Science, Physics) with a focus on biomedical imaging, causal inference, and AI for healthcare applications.

Software Engineer — Feb. 2025 - July 2025
Codegen International (Pvt.) Ltd. — *Colombo, Sri Lanka*

- Developed enterprise-grade applications with a strong foundation in software engineering principles, ensuring scalability and maintainability.
- Proficient in programming languages including Python, Java, and C++.
- Built and fine-tuned Large Language Models (LLMs) using techniques such as instruction tuning and parameter-efficient training (e.g., LoRA) for domain-specific tasks.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using vector databases to enable context-aware LLM responses.
- Deployed and evaluated LLM-based applications with focus on prompt engineering, hallucination reduction, and integration with real-world enterprise systems.

Trainee Software Engineer — Nov. 2024 - Feb. 2025
Codegen International (Pvt.) Ltd. — *Colombo, Sri Lanka*

- Actively contributing to projects involving AI/ML applications, including natural language processing (NLP) and aspect-based sentiment analysis.
- Gaining expertise in data preprocessing, model optimization, and evaluation on real-world datasets, including SemEval and domain-specific datasets.
- Enhancing proficiency in developing and deploying scalable machine learning solutions in cloud environments.

- Benefiting from a collaborative work environment and mentorship, furthering research and development in NLP and sentiment analysis technologies.

Student Instructor Jan. 2024 - Nov. 2024

Faculty of Engineering - University of Ruhuna *Galle, SriLanka*

- Conducted electronics and telecommunications laboratory sessions for undergraduate students, offering hands-on guidance to reinforce theoretical concepts.
- Assessed student performance and provided constructive feedback to support their academic growth and development.
- Designed and effectively implemented engaging experiments to enhance student understanding and practical skills in the field.

Research

Interests

- Applied Machine Learning, Artificial Intelligence, Natural Language Processing, Computer Vision, AI for Biomedical Applications

Bachelor's thesis

- Domain-Specific Sentiment Analysis in Textual Feedback: AI-Powered Aspect Detection and Sentiment Analysis for Textual Feedback and Reviews - In the digital age, customer feedback and online reviews are essential for business insights, fostering competitive advantages and customer loyalty. Aspect-Based Sentiment Analysis (ABSA), an advanced form of sentiment analysis, identifies and evaluates sentiments tied to specific text aspects, enhancing understanding of public opinions. Despite the progress enabled by AI, ABSA faces challenges with complex languages. This project aims to refine sentiment analysis to improve decision-making for various entities, leading to enhanced services and informed decisions.

PhD Research (Current Projects)

- Optical Neural Network Designs: Contributing to research on compact and efficient optical neural network architectures for next-generation AI-driven perception and computation.
- Out-of-Distribution (OoD) Detection: Exploring hypothesis-driven and causal inference methods to improve robustness and trustworthiness of deep learning models in biomedical imaging, computer vision, and natural language processing (NLP).
- Causal Inference for AI: Applying causal inference techniques across biomedical imaging, computer vision, and NLP to enhance interpretability, reliability, and fairness of machine learning models.
- Gaussian Splatting in Computer Vision: Investigating Gaussian splatting and related techniques for novel scene representations, rendering, and robust perception tasks.
- Natural Language Processing (NLP) and Large Language Models (LLMs): Researching domain-specific NLP applications including Aspect-Based Sentiment Analysis (ABSA), instruction-tuned LLMs, and retrieval-augmented generation (RAG) pipelines.

Publications

Jayakody, D.S., Isuranda, K., de Silva, N., Sandamali, G.G.N., Sudheera, K.L.K. and Rajith, S., "Enhanced Aspect-Based Sentiment Analysis with Integrated Category Extraction for Instruct-DeBERTa"
Accepted and presented - 38th annual meeting of Pacific Asia Conference on Language (PACLIC), Information and Computation, Tokyo, Japan, 2024

Jayakody, D.S., Isuranda, K., de Silva, N., Sandamali, G.G.N., Sudheera, K.L.K. and Rajith, S., Instruct-DeBERTa: A Hybrid Approach for Enhanced Aspect-Based Sentiment Analysis with Category Extraction. In Eighth Widening NLP Workshop (WiNLP 2024) Phase II.
Accepted - Eighth Widening NLP Workshop (WiNLP 2024) Phase II, EMNLP, Florida, USA, 2024

Jayakody, D.S., Isuranda, K., Malkith, A.V.A., De Silva, N., Ponnamperuma, S.R., Sandamali, G.G.N. and Sudheera, K.L.K., 2024, August. Aspect-based Sentiment Analysis Techniques: A Comparative Study. In 2024 Moratuwa Engineering Research Conference (MERCon) (pp. 205-210). IEEE.
Accepted and presented - MERCon 2024, 10th International Conference organized by the Engineering Research Unit, University of Moratuwa, Sri Lanka.

Jayakody, D.S., Malkith, A.V.A., Isuranda, K., Thenuwara, V., de Silva, N., Ponnamperuma, S.R., Sandamali, G.G.N. and Sudheera, K.L.K., 2024. "Instruct-DeBERTa: A Hybrid Approach for Aspect-based Sentiment Analysis on Textual Reviews."
Accepted - The International Journal on Advances in ICT for Emerging Regions (ICTer), 2024

Isuranda, K., **Jayakody, D.S.**, Malkith, A.V.A., "MCQ Automated Grading Using Image Processing Techniques"
Accepted - YMS Technical Conference 2024, organized by the Institute of Engineers Association, SriLanka

Ganegoda, G.S.S.S., **Jayakody, D.S**., Isuranda, M.A.K. and Priyankara, G., 2024. "GradesGo-A Comprehensive Practical Framework for Calculating Grade One Admission Marks."
Accepted and presented - Academic Sessions and Vice Chancellor's Awards 2024, University of Ruhuna

Main Projects

Domain-Specific Sentiment Analysis in Textual Feedback: AI-Powered Aspect Detection and Sentiment Analysis for Textual Feedback and Reviews
final year project

Student Formula EV Car - Ruhuna Racing - Manage the Electrical and Electronics Team
Manage the electrical & electronic team of the Ruhuna Racing project, guiding component integration for formula electric vehicles, manage integration of inverters, embedded systems, motors, and batteries for optimized performance.

Projects on AI for health
Developed convolutional neural networks for diagnosing lung and brain disorders and built risk models and survival estimators for heart disease prognosis using statistical methods and random forest predictors, created treatment effect predictors, applied model interpretation techniques, and used natural language processing to extract information from radiology reports.

Positions Held

IEEE Communications Society, University of Ruhuna
Vice Chairman *2023*

IEEE Robotics and Automation Society, University of Ruhuna
Event Organizer *2022*

Sports Council, Faculty of Engineering , University of Ruhuna
Vice President *2022*

University Swimming Team, University of Ruhuna
Captain *2023*

Awards, Competitions and Sports

Ronnie De Mel Gold Medal
Awarded for the best Engineering Graduate who obtained the highest Overall Grade Point in the whole batch with the First Class Honours.

Dr. A.D.V. Premaratne Memorial Gold Medal
Awarded for the Best Engineering Graduate who obtained the highest Overall Grade Point Average in the Department of Electrical and Information Engineering

BOC Award Gold Medal
Presented to the best overall student in the batch

Dean's Award for the Best Overall Performance in the batch
Presented to the best overall student in the batch in academics and extracurricular activities in 2021-2022, 2023-2024

HaXtreme 1.0 organized by the student chapters of ComSoc and Computer Society of IEEE student Branch of University of Ruhuna, 2022.
Member of the first place winning team

IEEE Xtreme, 2022
Member of the highest ranked team of University of Ruhuna

International Level Swimmer, 2010-2023
Represented Sri Lanka at the Junior World Aquatic Championship Singapore, vice captain of the Sri Lanka schools swimming team,National gold medalist, Sri Lanka university record holder, Sri Lanka schools record holder

Volunteering

Member of the AIESEC, University of Ruhuna

Member of the Rotaract Club, University of Ruhuna

Engaging as a volunteer teacher in the 'Mehewara' project, organized by the Faculty of Engineering, University of Ruhuna, to enhance the mathematics knowledge of ordinary level students in underprivileged rural areas

Referees

Assistant Prof. Dushan Wadduwage : dwadduwa@odu.edu
Ph.D. (Singapore-MIT Alliance Graduate Fellowship), Biological and Biomedical Sciences, B.Sc. Eng. (Hons) in Electronics and Telecommunication Engineering (University of Moratuwa),Assistant Professor (Tenure-Track), Department of Data Science, Computer Science, and Physics, Old Dominion University, USA

Dr. Chatura Seneviratne : chatura@eie.ruh.ac.lk
PhD (Calgary, Canada), M.Eng. (AIT, Thailand), B.Sc. Eng. (Hons) (Ruhuna), Senior Lecturer, Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna, Sri Lanka

Dr. Kushan Sudheera : kushan@eie.ruh.ac.lk
PhD (NTU, Singapore), BSc. Eng. (Hons)(Ruhuna), MIEEE, AMIE (SL) , Senior Lecturer, Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna, Sri Lanka

Dr. Sujith Wijerathne : sujith_wijerathne@nuhs.edu.sg
MBBS (Sri Lanka), MRCS (Ed), MMed (Surgery), FRCS (Ed), FAMS (General Surgery), EMBA (NUS), EMBA (UCLA), Adjunct Associate Professor, Department of Surgery, Yong Loo Lin School of Medicine, NUS, Sinagapore